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Classification and Regression
Report

Logistic Regression prediction results :

Training set Accuracy for BLR : 92.672%
Validation set Accuracy for BLR : 91.5%
Testing set Accuracy for BLR : 89.04%

The higher training accuracy (92.672%) compared to the testing accuracy (89.04%) suggests that the model might be slightly overfitting the training data. This means the model has learned the training data too well, including some of its noise and peculiarities, which don't generalize to the test set.

Multiclass - Logistic Regression prediction results :

Training set Accuracy for MLR:93.448%
Validation set Accuracy for MLR:92.47%
Testing set Accuracy for MLR:92.55%

MLR performs better as it captures the nuances and relationships between multiple categories simultaneously, leading to improved accuracy.

Comparison between performances of Linear kernel and Radial basis :

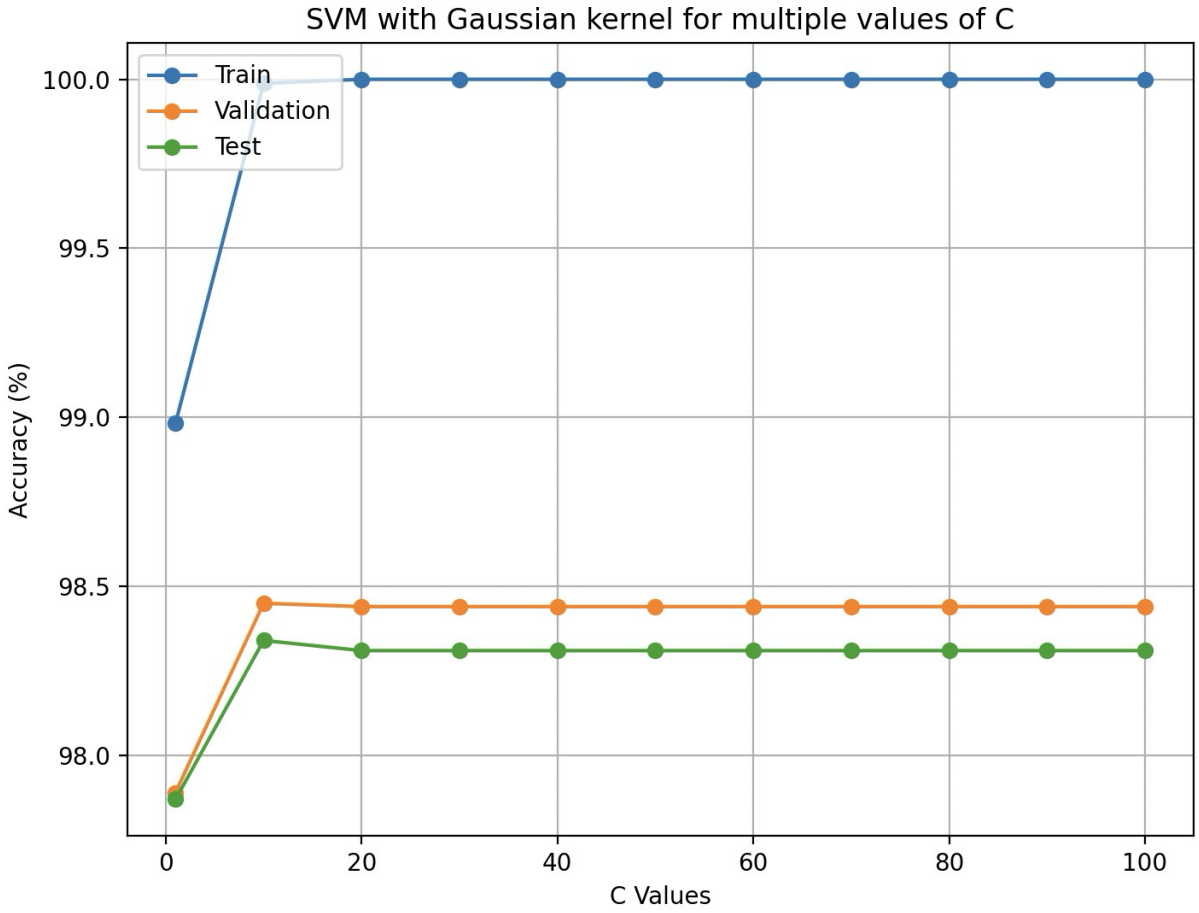
Kernel	Gamma	Training	Validation	Testing
Linear	**0**	**97.286%**	**93.64%**	**93.78%**
Radial basis	**0**	**98.982%**	**97.89%**	**97.87%**
Radial basis	**1**	**100.0%**	**15.479999%**	**17.14%**

Comparing kernel performance, the RBF kernel with gamma value of 0 achieves a 94.42% accuracy, performing marginally better than the Linear kernel's 93.78% accuracy. However, while the RBF kernel with gamma value of 1 achieves perfect accuracy (100%) on the training dataset, it exhibits poor performance on both validation and test sets, which may be due to overfitting.

Using radial basis function with value of gamma setting to default and varying value of C :

C	Training Accuracy	Validation Accuracy	Testing Accuracy
1.0	98.982%	97.89%	97.87%
10.0	99.988%	98.45%	98.34%
20.0	100.0%	98.44%	98.31%
30.0	100.0%	98.44%	98.31%
40.0	100.0%	98.44%	98.31%
50.0	100.0%	98.44%	98.31%
60.0	100.0%	98.44%	98.31%

70.0	100.0%	98.44%	98.31%
80.0	100.0%	98.44%	98.31%
90.0	100.0%	98.44%	98.31%
100.0	100.0%	98.44%	98.31%



As C increases, the accuracy improves. With a lower C value, each error term carries less weight, resulting in a larger margin hyperplane. Conversely, when C is set higher, the error terms carry more weight, leading to a smaller margin hyperplane and increased accuracy.
However, setting C too high leads to overfitting as we can see from the graph, while setting it too low can result in underfitting. Therefore, careful selection of the C value is essential for optimal model performance.